Indian Institute of Technology, Madras

Computational Neuroscience (BT6270)

Assignment 3

Name: Aaradhy Siorthia
Roll No: Na18b008

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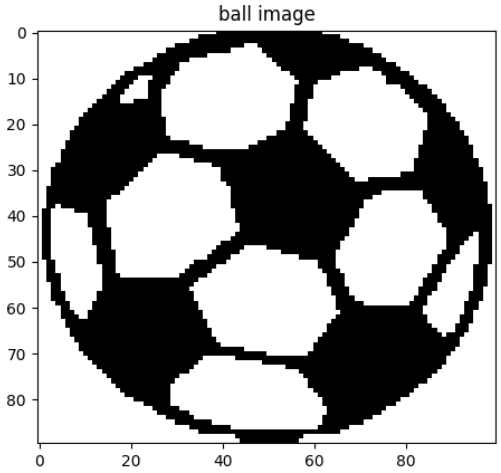
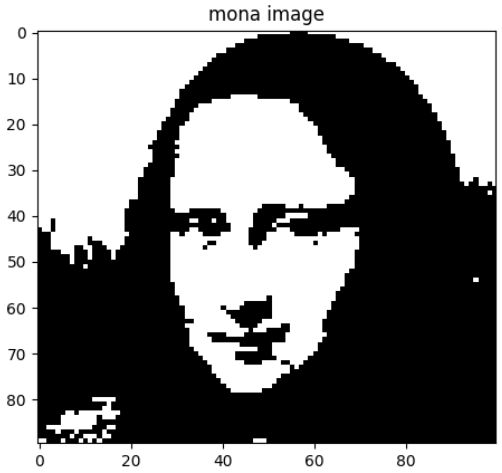
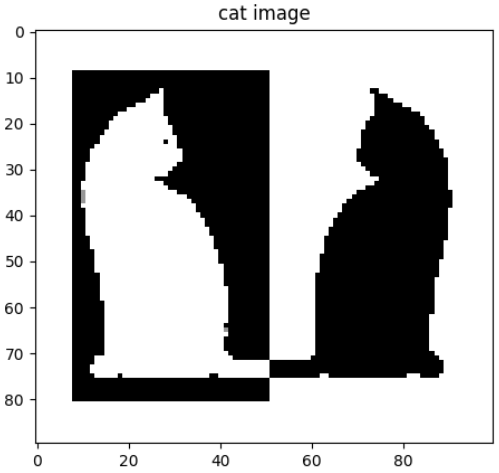
1. Three figures (mona, ball, cat) are given in .txt format. Each figure is a 90x100 matrix.

a. Visualize the images and make sure that the black pixels are represented by -1 and white pixels are represented by +1.

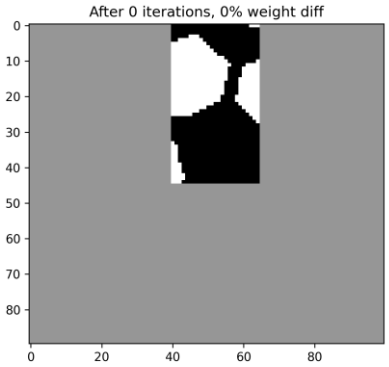


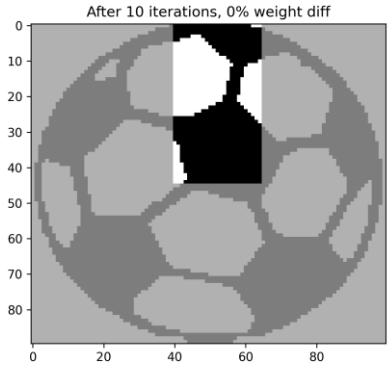


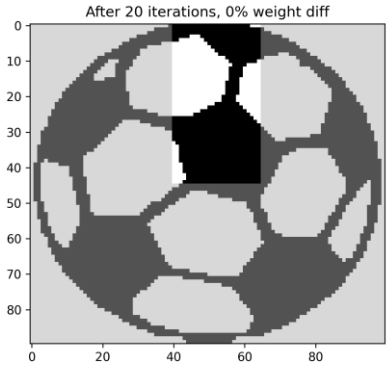


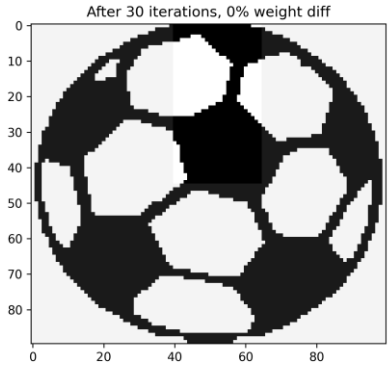
2. Ball Portion Plotting

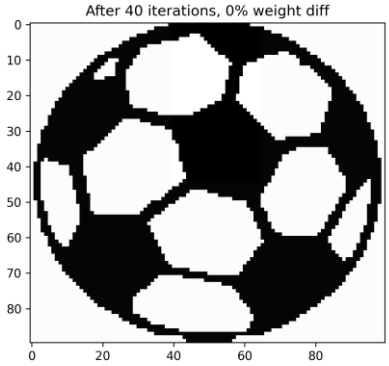
Football image is stored in the Network. Now from patch of original image will retrieve the whole image. Patch of image as an INPUT for network:

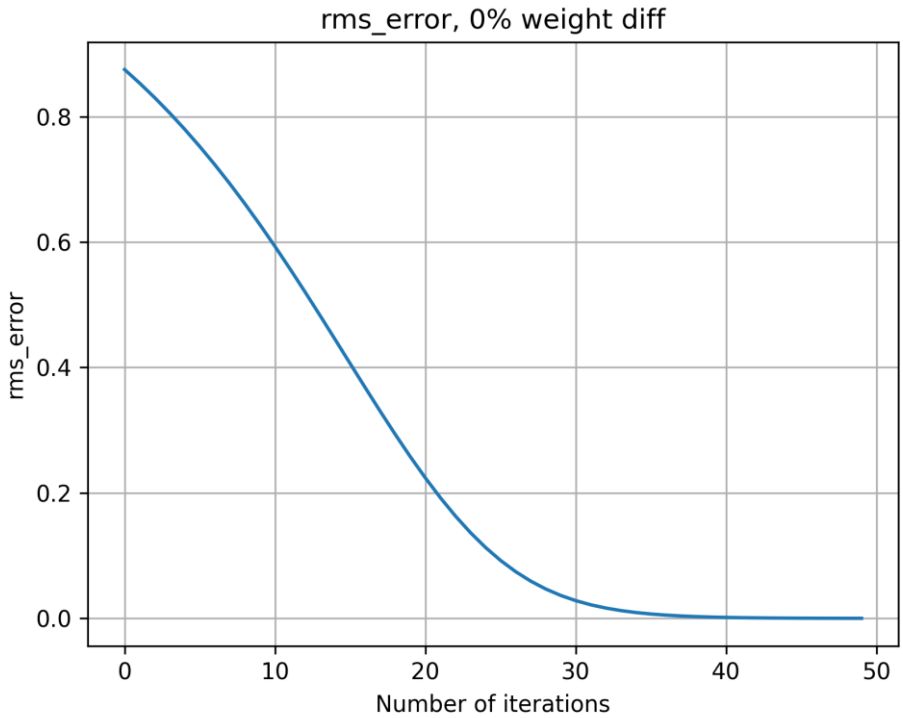










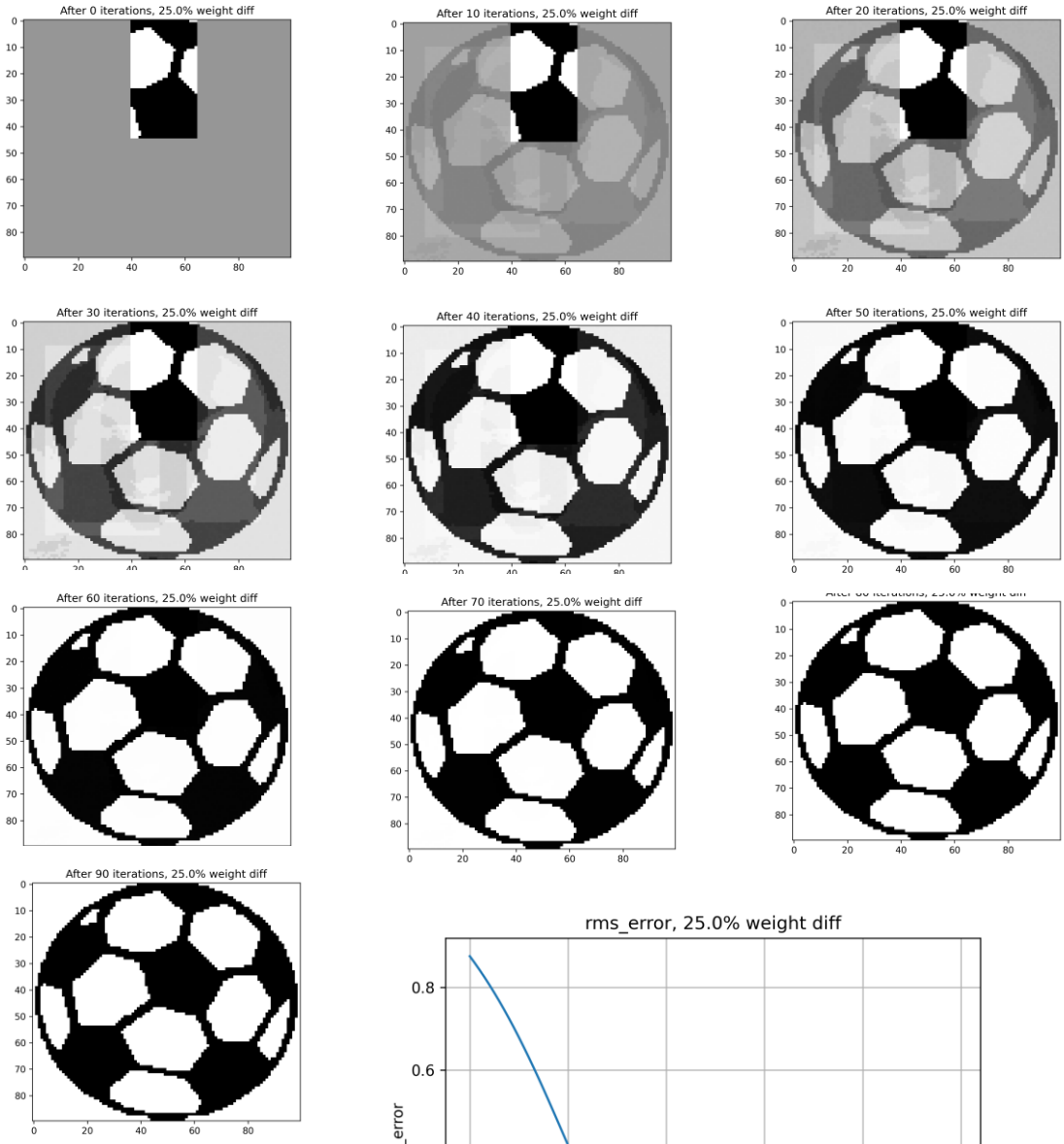


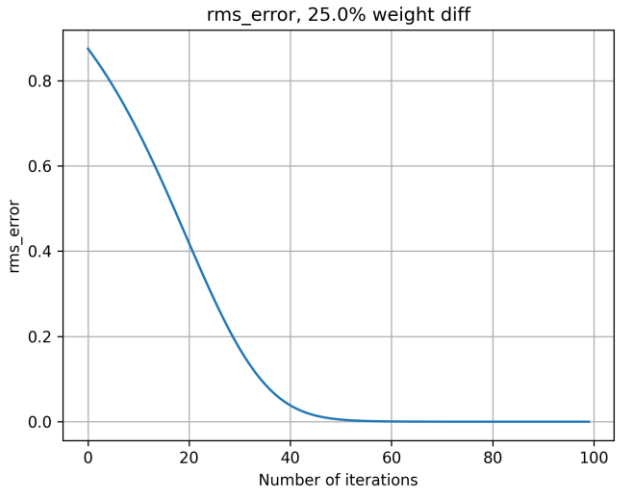
rms_E vs number of iteration plot shows that after some iteration rms_E decreases exponentially

3. Image regeneration

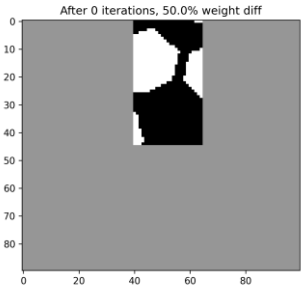
a. Ball

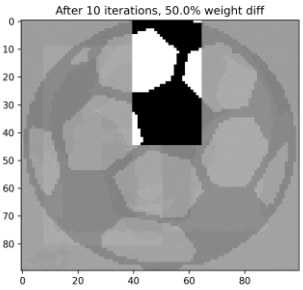
i. 25% weight difference

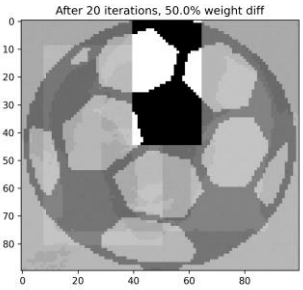


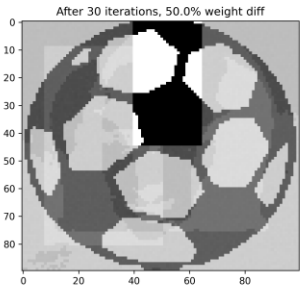


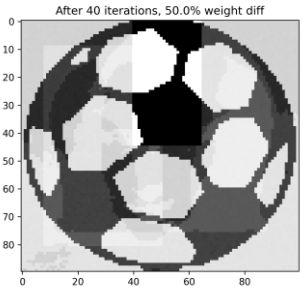
ii. 50% weight difference

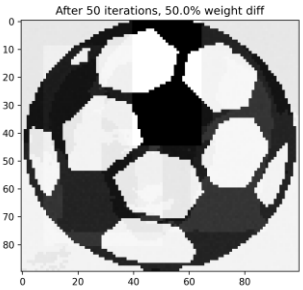


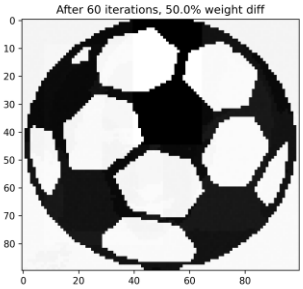


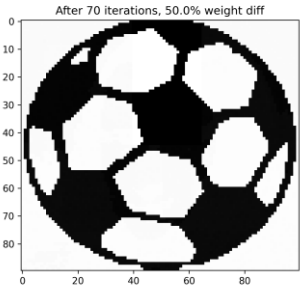


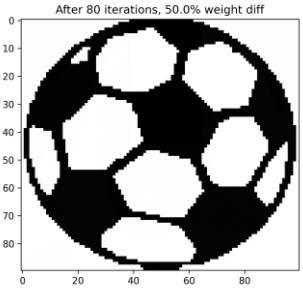


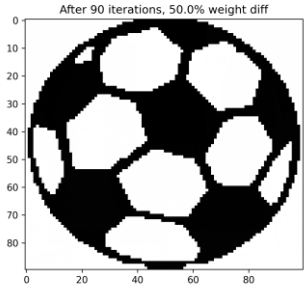


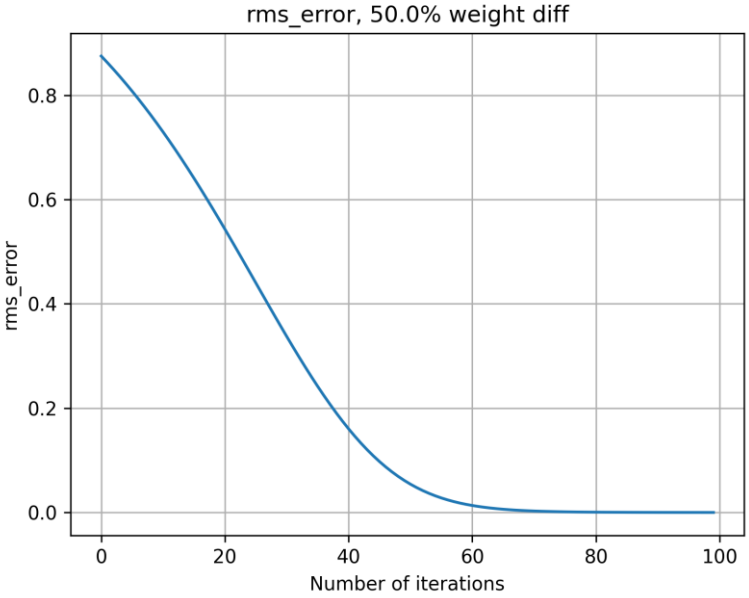




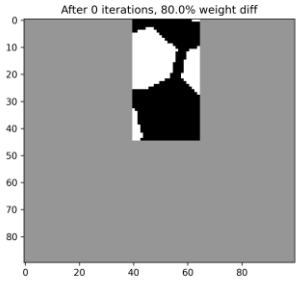


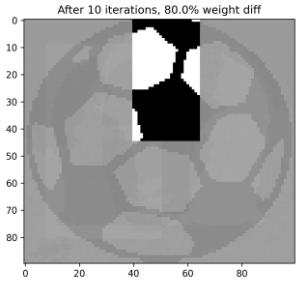


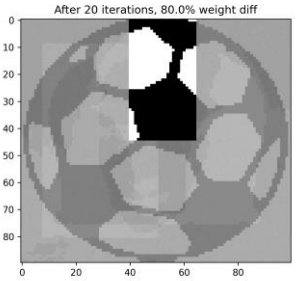


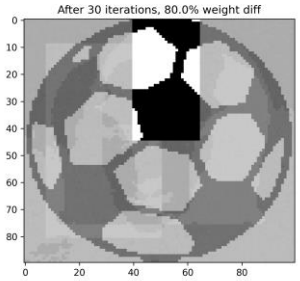


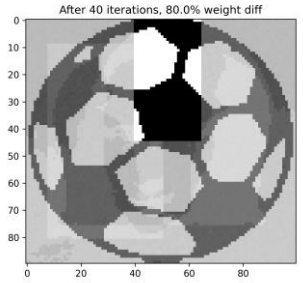
iii. 80% weight difference

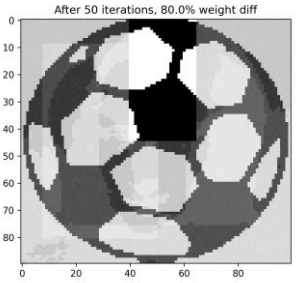


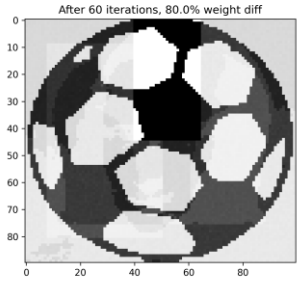


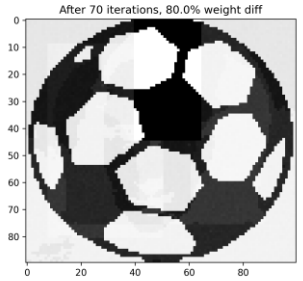


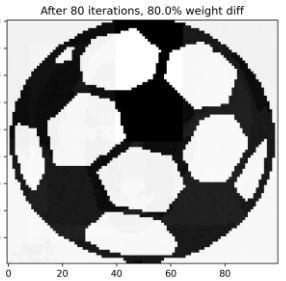


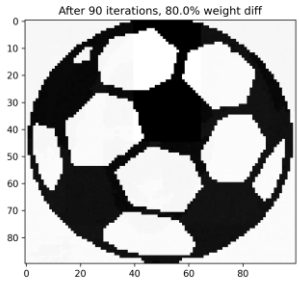


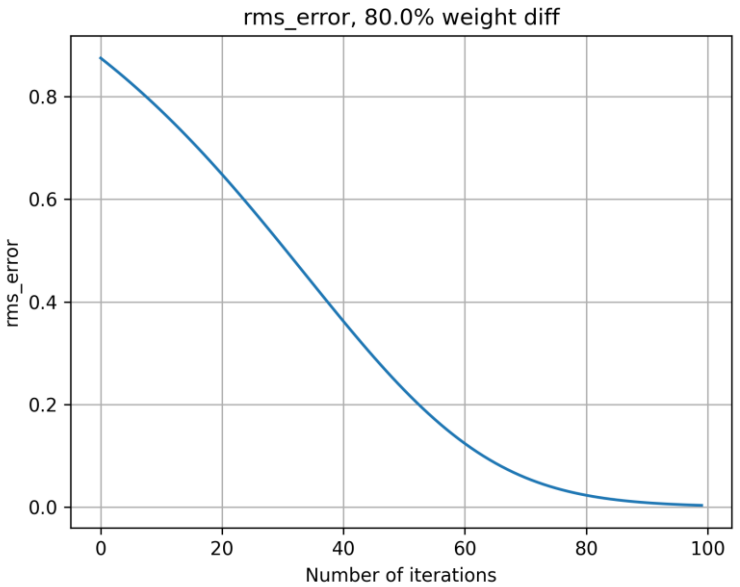






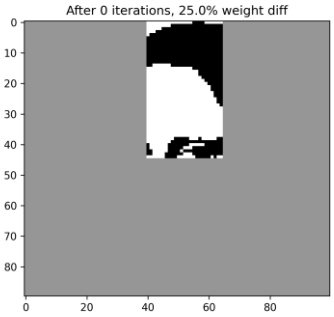


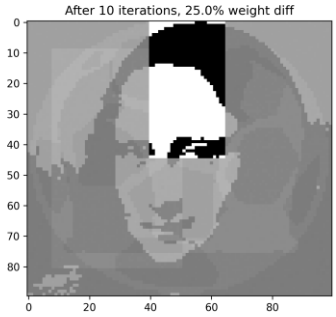


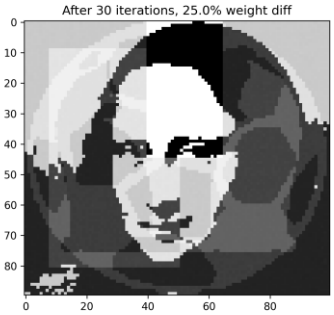


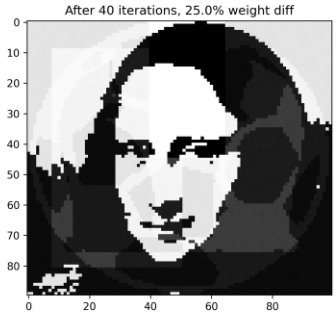
b. Mona

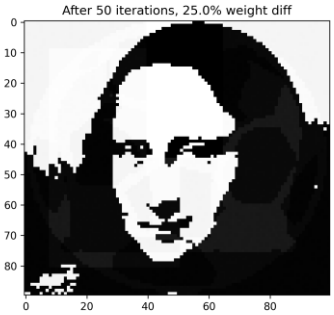
i. 25% weight difference

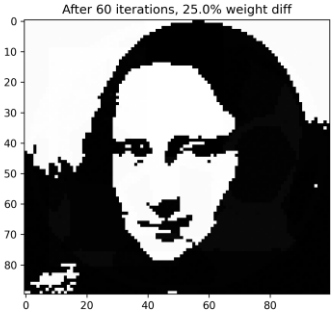


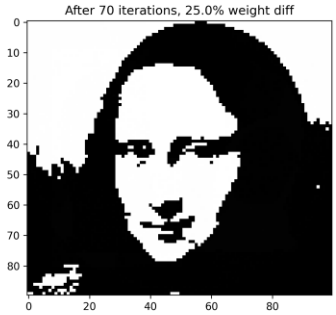


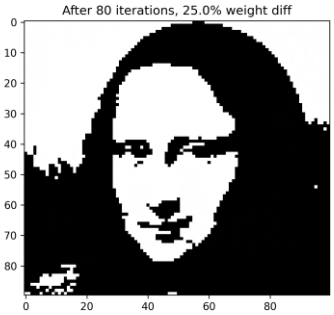


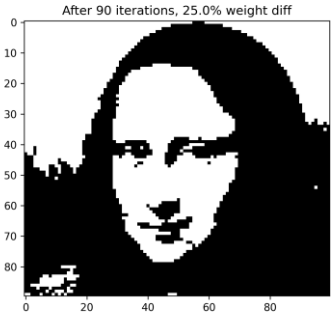


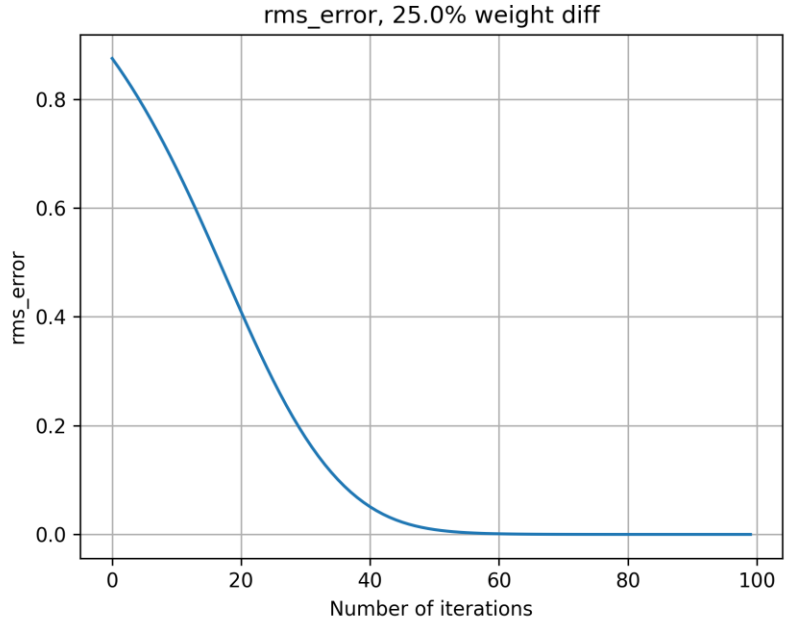




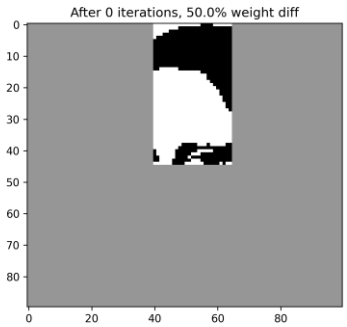


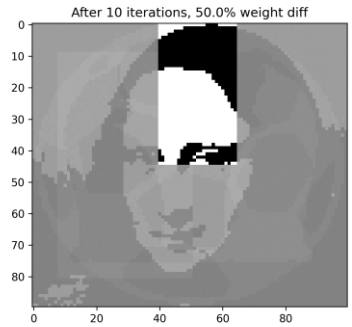


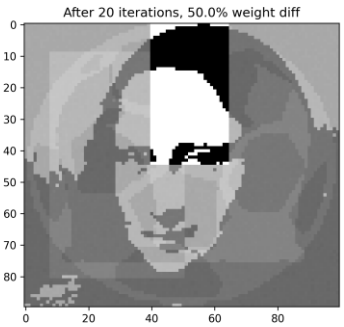


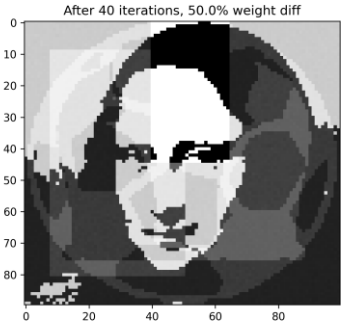


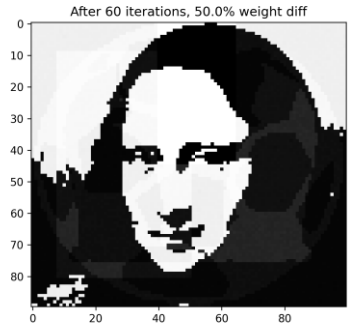
ii. 50% weight difference

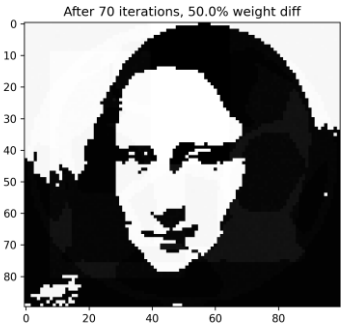


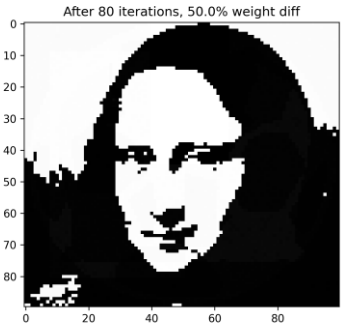


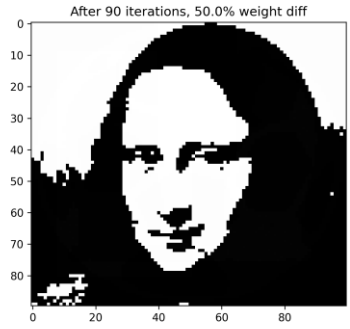


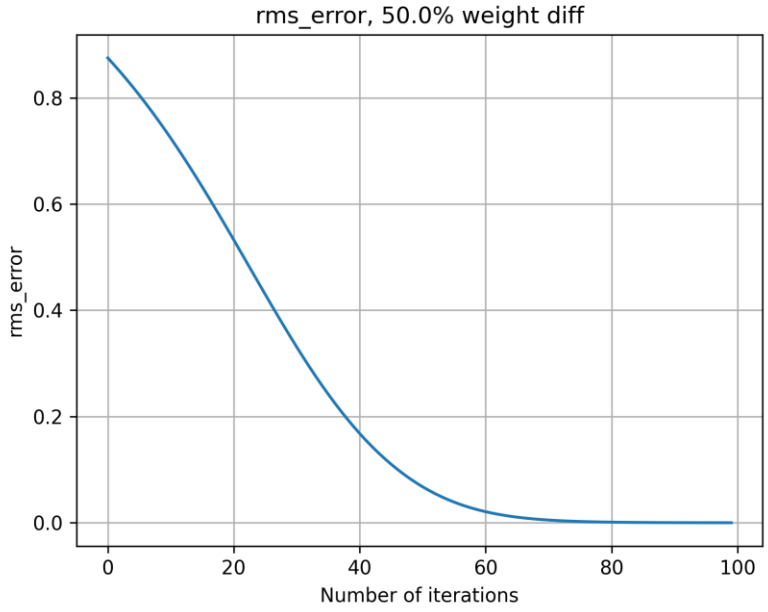




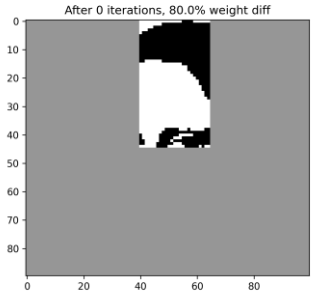


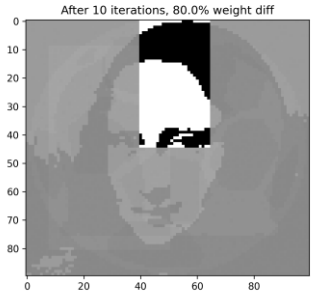


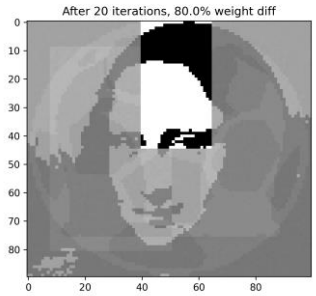


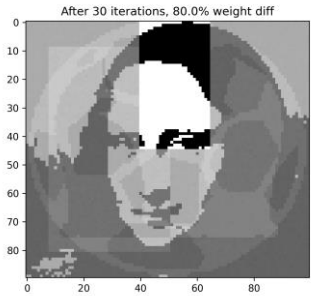


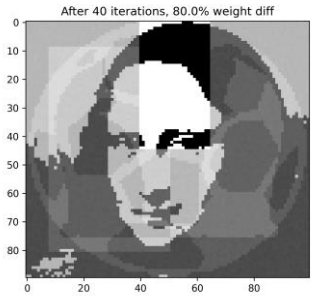
iii. 80% weight difference

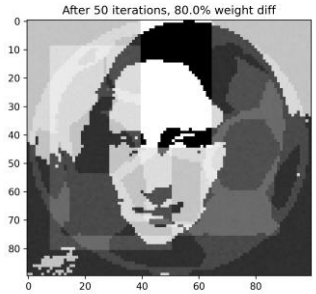


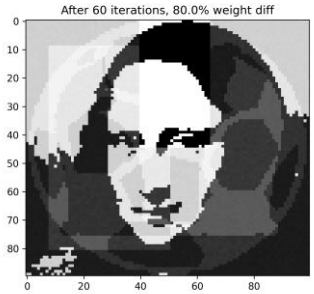


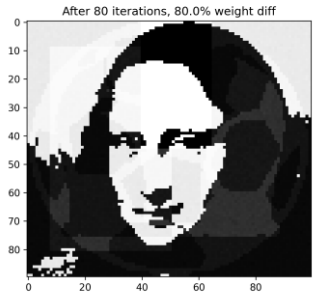


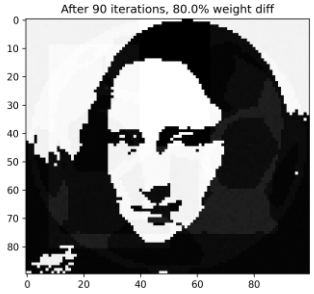


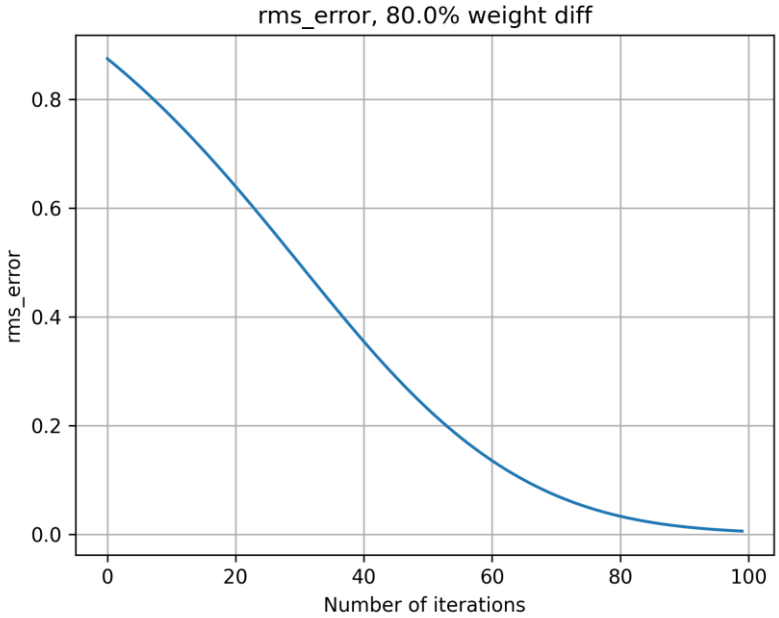






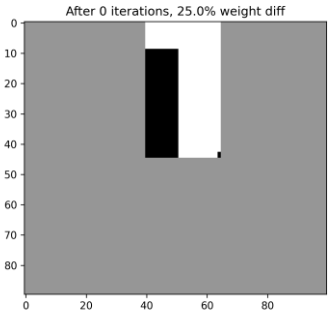


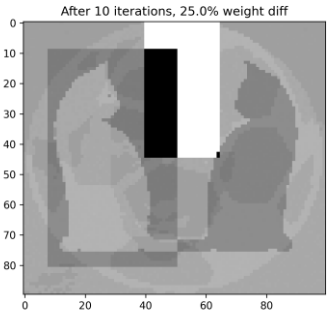


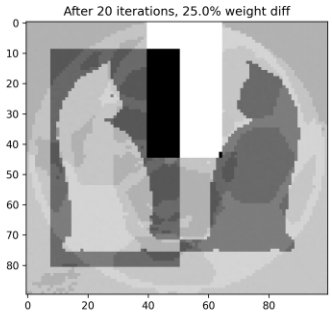


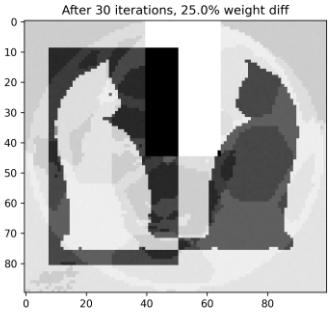
c. Cat

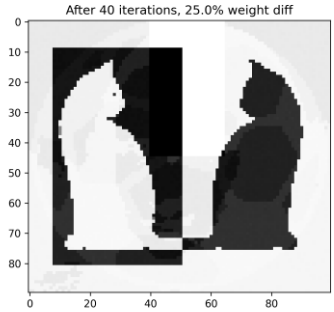
i. 25% weight difference

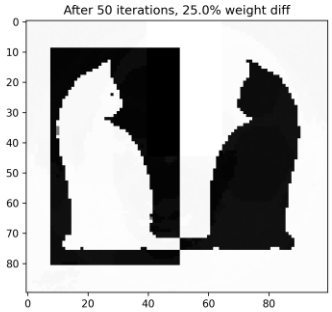


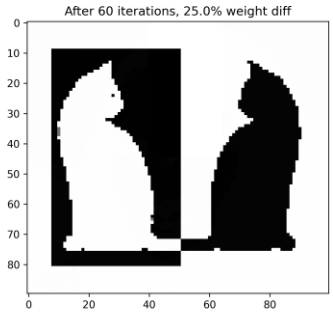


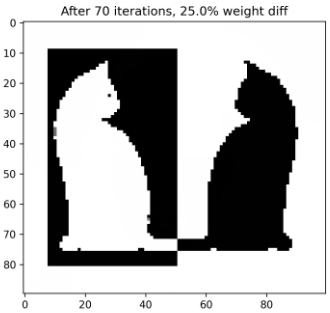


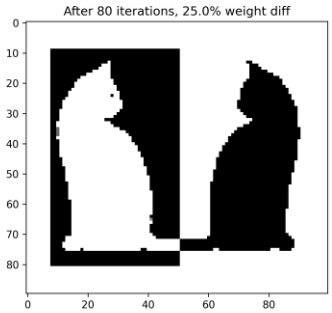


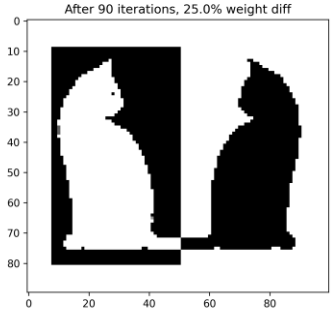


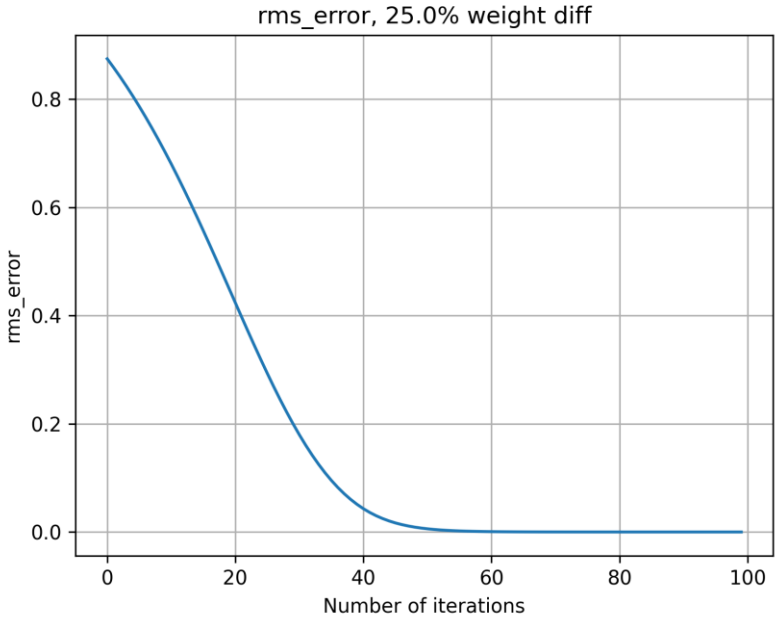




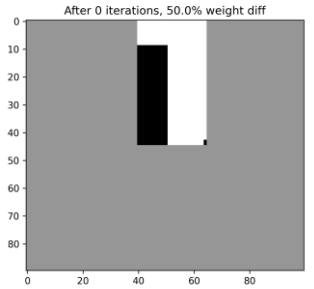


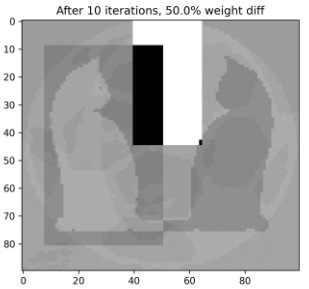


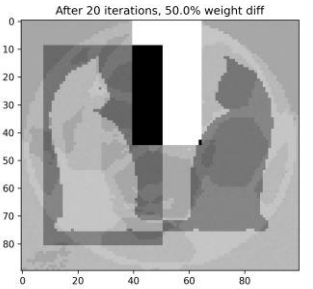


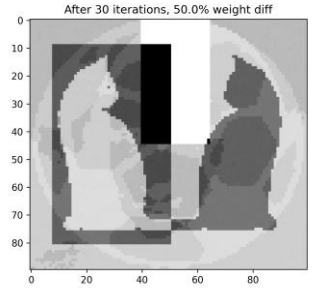


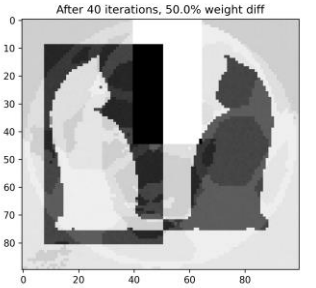
ii. 50% weight difference

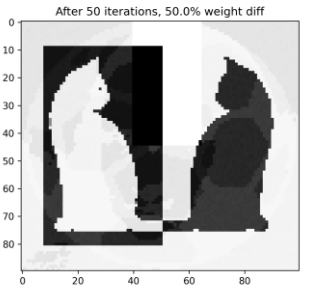


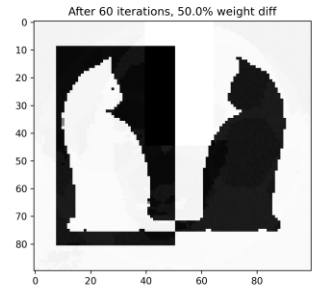


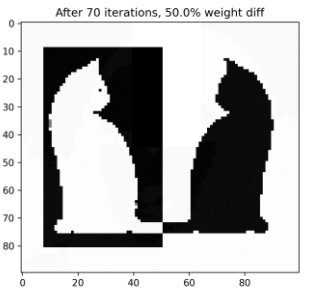


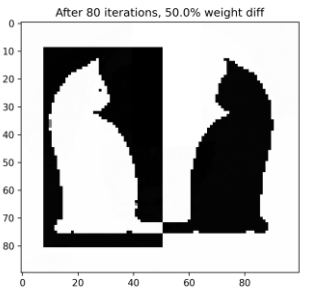


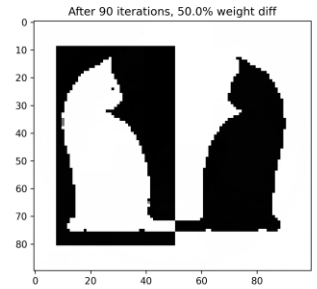


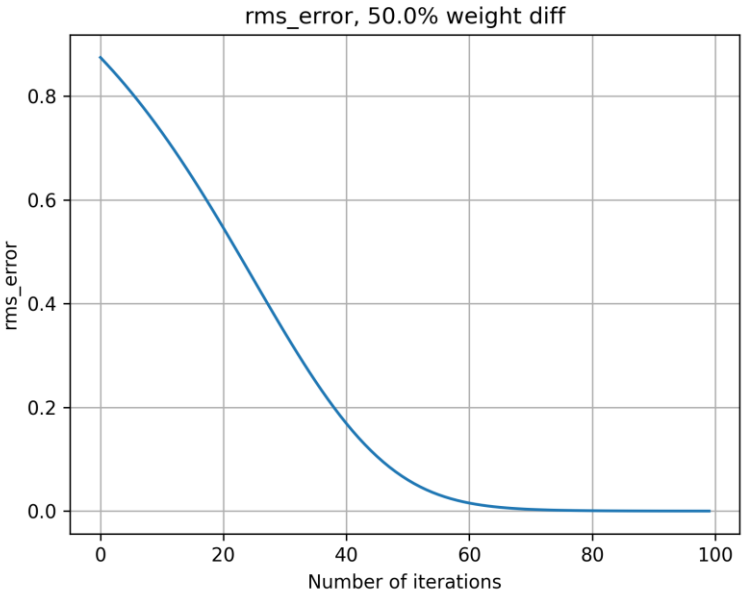
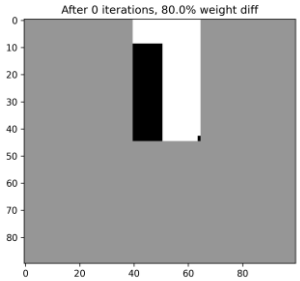
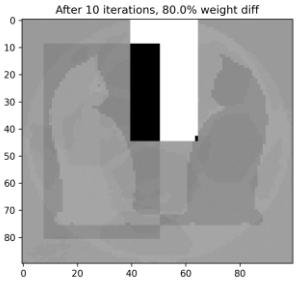
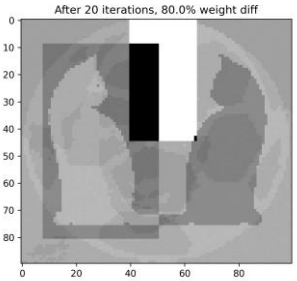
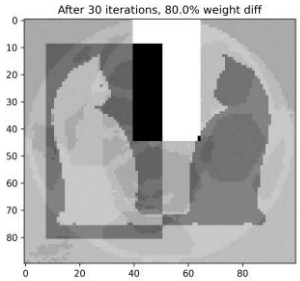
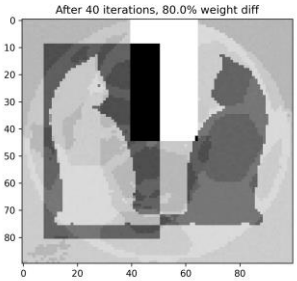
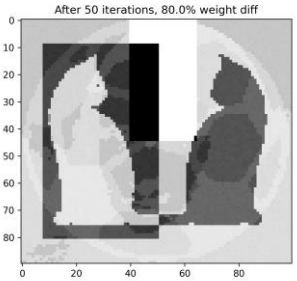
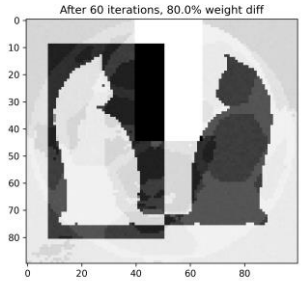
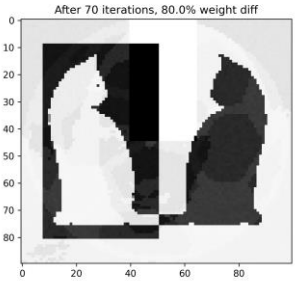
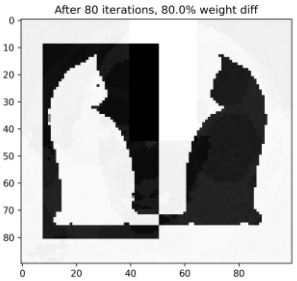
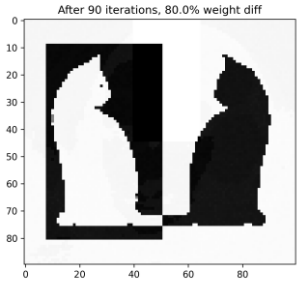
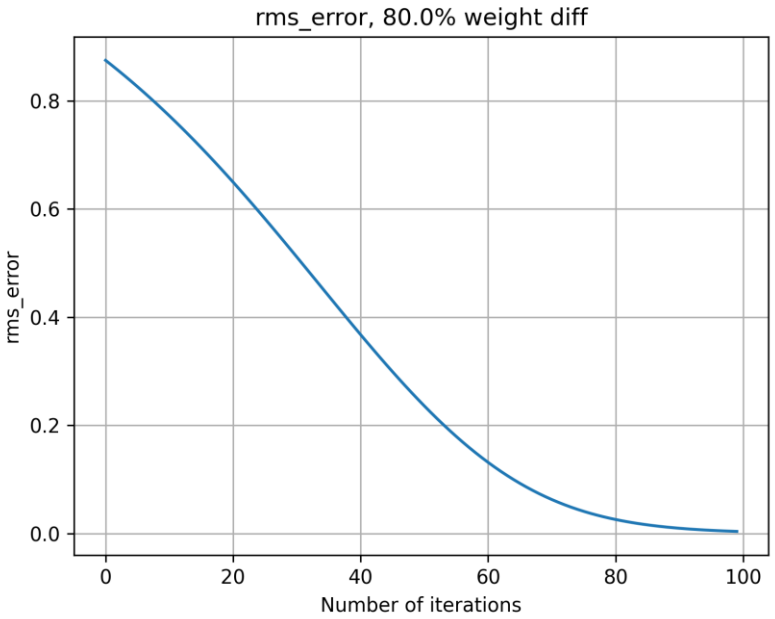












iii. 80% weight difference